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Kinematic and Dynamic Analysis of Escapement Mechanism

ME444: Analysis and Design of Mechanical Systems
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Abstract

Mechanical clocks have been central to human timekeeping for centuries, and at the heart of every mechanical clock lies an escapement mechanism. This project set out to study and understand how escapements work, eventually focusing on the deadbeat escapement, a mechanism known for its accuracy and zero-recoil behavior, making it particularly interesting for kinematic and dynamic analysis.

We approached the project through three parallel tracks. We began with theoretical analysis, deriving the pendulum period, calculating gear train ratios, and studying the escapement geometry to understand how each component contributes to regulated timekeeping. Alongside this, we built a functional clock prototype using 3D printed components, which gave us a physical platform to observe and validate the mechanism in action. We also developed a Python-based simulation to model pendulum motion and animate the escapement, producing displacement-time plots and phase portraits to study the oscillation behavior.



Figure 1: clock prototype

Through the combination of theory, prototype, and simulation, we were able to study the deadbeat escapement from multiple angles, understanding not just how it works on paper, but why it works so well in practice. The project highlights how classical mechanism analysis comes to life when connected to a real, functioning system.

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1 Introduction

1.1 Background

The development of the **mechanical clock** was a major step in how people understood and measured time. Earlier methods, like water clocks or candle markings, relied on natural processes that were not always consistent or easy to control. Mechanical clocks improved this by using systems of gears, wheels, and stored energy to keep time in a more reliable way.
This change was not just about convenience. Accurate timekeeping became important for organizing daily life, coordinating trade and markets, and supporting scientific work. It was especially useful in astronomy, where precise timing is needed to study the motion of celestial bodies. Most importantly, it helped solve the longitude problem, making long-distance sea travel much safer.

At the core of this system is the **escapement**, often called the *"heartbeat"* of the clock. This mechanism controls how energy is released. Instead of letting all the stored energy from a weight or spring run out at once, it releases it in small, controlled steps. At the same time, it gives a small push to the pendulum to make up for energy lost due to friction and air resistance. Without an escapement, a clock cannot run at a steady rate or convert mechanical energy into accurate time measurement.

1.1.1 Deadbeat escapement

The evolution of escapement design has been a centuries-long effort to minimize interference between the power source and the timekeeping element. Early designs such as the *verge escapement* were limited by high friction and large oscillations. The later *anchor escapement* improved efficiency but introduced **recoil**, a backward motion of the gear train that increased wear and reduced accuracy.

These limitations were addressed in 1715 by **George Graham**, who perfected the **deadbeat escapement**. By designing the pallet surfaces to be concentric with the pivot point, Graham eliminated recoil entirely. This ensured that the escape wheel remained stationary, or "dead," during the pendulum's swing.

The introduction of the deadbeat escapement marked a major advancement in mechanical design:

- **Elimination of Recoil:** The removal of backward motion significantly reduced mechanical wear and improved durability.
- **Precision and Stability:** The pendulum motion became less disturbed, making the clock's rate less sensitive to variations in the driving force such as lubrication changes or spring tension.

In this project, we aim to perform a detailed **kinematic and dynamic analysis** of this mechanism. By studying the transition from recoil-based systems to the deadbeat escapement, we seek to analyze the geometric constraints that allow for zero-recoil behavior. Our approach combines theoretical derivations with a **Python-based simulation** and a **3D-printed physical prototype** to validate how Graham's 18th-century design continues to serve as a masterclass in mechanical regulation and precision.

1.2 Literature Survey

1.2.1 Evolution of Clock Escapement Mechanisms

Stoimenov, Popkonstantinović, Miladinović, and Petrović
FME Transactions, 2012[1]

This paper presents a comprehensive study of the evolution of clock escapement mechanisms over seven centuries, including the crown wheel and verge, anchor recoil, deadbeat, and detached escapements. For each mechanism, the authors analyze the geometric design, kinematic behavior, and dynamic interaction between the escapement and the pendulum using moment interaction diagrams to evaluate timekeeping accuracy.

A key contribution relevant to this project is the distinction between recoil and deadbeat escapements. In the recoil anchor escapement, both the impulse and locking functions occur on the same pallet surface. As a result, the escape wheel experiences a slight backward motion during the locking phase. This leads to energy loss and disturbs the pendulum motion, producing *tachychronous* behavior, where an increase in driving force results in a faster clock rate.

In contrast, the deadbeat escapement separates the locking and impulse surfaces. The locking surface is designed as a circular arc centered at the pallet pivot, which eliminates the moment of the locking force at the suspension point. This completely removes recoil and results in *brachychronous* behavior, making the

clock significantly less sensitive to variations in driving force. The mechanism achieves a daily rate error of approximately ±3 to 5 seconds.

This study provided a clear theoretical basis for selecting the deadbeat escapement for our analysis and guided our understanding of the locking and impulse phases in the mechanism.

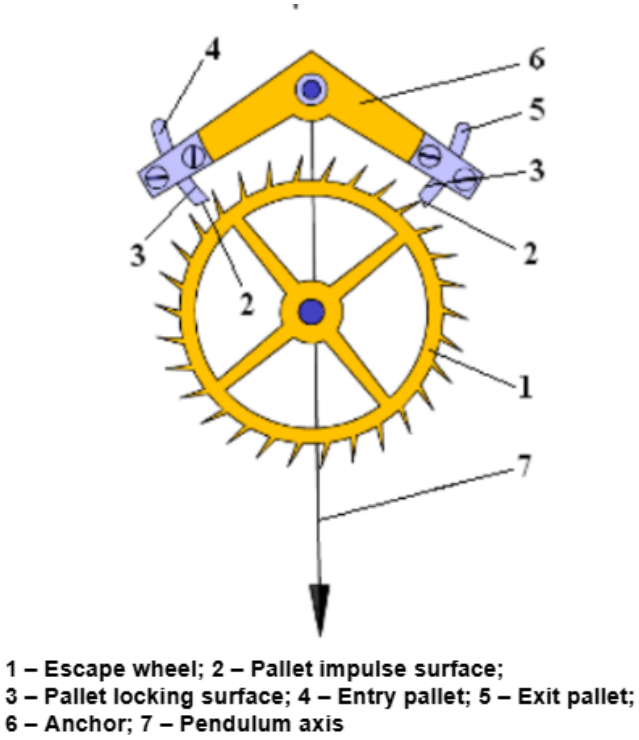


Figure 2: Deadbeat escapement mechanism showing escape wheel, entry and exit pallets, and locking and impulse surfaces [1]

1.2.2 Model of a Mechanical Clock Escapement

Moline, Wagner, and Volk
American Journal of Physics, 2012[2]

This paper develops a nonlinear mathematical model of the Graham deadbeat escapement using a real Seth Thomas tower clock at Clemson University. The pendulum, crutch, and verge are modeled as a single rotating system, and the equations of motion are derived using Haug's method. Contact forces between the escape wheel teeth and pallet surfaces are modeled using linear stiffness and damping, with friction included in both locking and impulse phases.

The model is experimentally validated using gyroscopes mounted on the verge

and pendulum, with the measured oscillation period matching the theoretical prediction within 0.7%.

Equations of Motion

The pendulum, crutch, and pallet fork are treated as a single rigid body rotating about a fixed pivot. The governing equation of motion is:

$$I\ddot{\theta} + c\dot{\theta} + mgL\sin\theta = \tau_{\text{contact}}(t) \tag{1}$$

where I is the combined moment of inertia of the pendulum-crutch-pallet assembly, c is the viscous damping coefficient capturing air resistance and bearing friction, mgL is the restoring gravitational torque, and $\tau_{\text{contact}}(t)$ is the time-varying contact torque from the escape wheel teeth acting on the pallet surfaces. For small oscillations ($\sin\theta \approx \theta$), this reduces to the standard damped harmonic oscillator with natural frequency $\omega_0 = \sqrt{mgL/I}$ and undamped period $T = 2\pi\sqrt{L/g}$.

Contact Force Model

The interaction between escape wheel teeth and pallet faces is modeled as a compliant contact using a linear spring-damper:

$$F_{\text{contact}} = k_c\,\delta + b_c\,\dot{\delta} \tag{2}$$

where δ is the instantaneous tooth penetration depth, and k_c, b_c are the contact stiffness and damping coefficients respectively. Coulomb friction is superimposed as $F_{\text{friction}} = \mu F_{\text{normal}}$, with distinct friction coefficients applied during the locking phase (sliding on the concentric circular arc faces) and the impulse phase (contact on the angled impulse faces).

Phase-Plane Analysis

A phase portrait of $\dot{\theta}$ versus θ reveals the limit cycle behavior of the pendulum. In a correctly tuned deadbeat escapement, the pendulum settles into a stable closed orbit in phase space, indicating a self-sustaining oscillation in which the energy supplied by the gear train exactly compensates for dissipative losses each half-cycle. The impulse phase manifests as a small outward perturbation in the phase trajectory at each half-swing, after which the orbit returns to its limit cycle, confirming amplitude stability.

Energy Budget per Cycle

The model enables a quantitative breakdown of energy dissipation per oscillation cycle. Losses occur through three primary channels:

- Viscous damping at the pendulum suspension ($\approx$ 60–70% of total dissipation).
- Sliding friction at the pallet-tooth interface during the locking phase ($\approx$ 15–20%).
- Impulsive friction during the tooth drop event ($\approx$ 10–15%).

The driving torque from the gear train must replenish these losses precisely each half-cycle; the impulse face geometry is designed to deliver the correct energy increment regardless of small variations in the driving force.

Sensitivity to Geometric Parameters

A significant finding of the study concerns the sensitivity of escapement performance to geometric tolerances. Two critical angles govern the pallet geometry: the locking angle α_3, which defines the angular extent of the concentric arc, and the impulse angle α_2, which defines the slope of the impulse face. Deviations as small as 0.01 mm in fabricated pallet dimensions alter the impulse torque magnitude and shift the pendulum's stable amplitude, potentially producing systematic gaining or losing behavior in the clock rate. This sensitivity explains the historical demand for precision horological craftsmanship and motivates the tight manufacturing tolerances required in our own 3D-printed prototype.

Several observations from this study are directly relevant to our project:

- During the locking phase, the escape wheel exhibits zero recoil and remains stationary.
- The escape wheel advances only during the impulse phase by approximately 9° per interaction.
- The pendulum demonstrates smooth harmonic motion with well-defined amplitude and period.
- Gear train analysis illustrates how torque from the driving weight is transmitted through multiple stages to the escape wheel, informing our own gear ratio calculations.

- Small geometric deviations on the order of 0.01 mm in pallet dimensions can significantly affect system behavior, emphasizing the need for precise fabrication.

This paper served as the primary technical reference for our theoretical modeling, pendulum dynamics analysis, and validation approach.

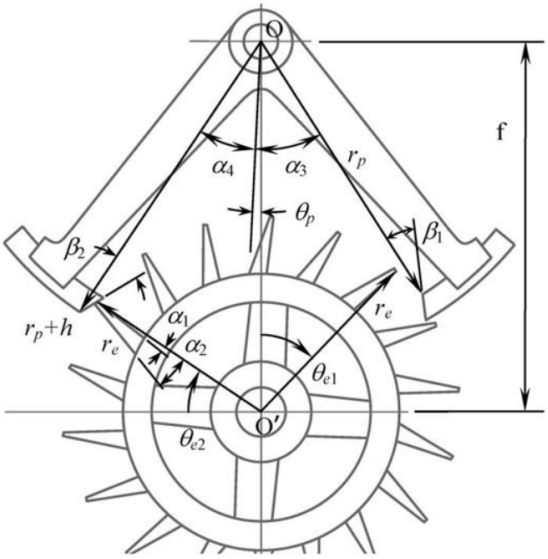


Figure 3: Geometry of escape wheel and pallet interaction illustrating twin pallet mechanism [2]

1.3 Motivation

Our interest in this project began with a fundamental curiosity about how mechanical systems can maintain high-precision consistency using nothing more than gravity and a series of gears. In an era dominated by digital and electronic clocks, we wanted to understand the "analogue" logic that allowed humanity to track time for centuries. We soon realized that the secret lies entirely in the escapement mechanism, which serves as a masterclass in how geometry can be used to regulate energy.

As students of ME444: Analysis and Design of Mechanical Systems, we were looking for a way to apply the theories we learned in class—kinematics, dynamics, and mechanism design—to a real-world machine. The deadbeat escapement is a perfect candidate for this because it involves:

- **Contact Mechanics:** Studying exactly how the teeth of the gear hit and slide against the pallets.
- **System Stability:** Understanding how to keep a pendulum swinging at a constant rate despite friction.
- **Precision Design:** Seeing how a specific shape can eliminate the "jerky" backward motion found in older clocks.

What made the project even more appealing was the possibility of building a fully working physical prototype. There is something genuinely satisfying about understanding a mechanism analytically and then watching it come to life in your hands. The availability of an open-source CAD model of a complete skeleton clock made this feasible within our project timeline.
Finally, the deadbeat escapement has a certain elegance to it—a mechanism designed in 1715 that solved a fundamental problem in mechanical design through pure geometry, and which continued to be used in precision clocks for centuries. Studying it felt less like a course requirement and more like uncovering something worth understanding properly.

1.4 Description of the Mechanism

The deadbeat escapement is a precision regulating mechanism designed to convert continuous rotational energy into controlled, intermittent steps. It consists of four primary components that work in synchronization to maintain the clock's frequency.

The Escape Wheel & Power Source
The system is driven by a gravity-fed weight that applies constant torque to the escape wheel through a gear train. In our design, the escape wheel features 30 teeth. Without the escapement's regulation, this wheel would spin uncontrollably; instead, it is permitted to advance by exactly 12° per tick, ensuring energy is released only as needed.

The Pallet Fork
The most critical engineering feature of this mechanism is the pallet fork. Unlike older designs, the "locking faces" of our pallets are circular arcs concentric to the pivot point. This specific geometry ensures that when a tooth hits the pallet, the force passes directly through the pivot, resulting in zero torque. This eliminates

"recoil," meaning the escape wheel remains perfectly still (or "dead") while the pendulum swings, significantly reducing mechanical wear.

The Pendulum
The timing is governed by a pendulum. Based on the formula

$$T = 2\pi\sqrt{\frac{L}{g}}$$

this length provides the consistent period required for timekeeping. As the pendulum swings, it moves the pallet fork, triggering a three-phase cycle:

- **Locking:** The wheel is held stationary by the concentric pallet face.
- **Impulse:** As the pallet slides open, the tooth hits an angled "impulse face," giving the pendulum a tiny mechanical push to overcome air resistance.
- **Drop:** The tooth leaves the pallet, and the wheel rotates freely for a split second before the opposite pallet catches the next tooth, creating the "tick" sound.

The Gear Train (Output)
Finally, the regulated motion of the escape wheel is transmitted back through a gear train. By using specific gear ratios, the high-frequency "ticks" are slowed down to drive the minute and hour hands at their respective speeds, completing the transition from raw potential energy to readable time.

1.5 Applications of the Mechanism

While the deadbeat escapement is most commonly associated with grandfather clocks and tower clocks, its underlying principles extend far beyond traditional timekeeping. The mechanism represents a broader engineering concept—using precise geometry to regulate energy release—which finds relevance in several modern and historical applications.

Precision Pendulum Clocks
The most direct application is in high-precision pendulum clocks used in observatories and scientific institutions. The deadbeat escapement was the standard regulator in astronomical clocks throughout the 18th and 19th centuries, where accurate timekeeping was essential for celestial observations and navigation.

Tower and Public Clocks

Large tower clocks, including many still in operation today, use deadbeat or gravity escapement mechanisms derived directly from Graham's original design. The Big Ben clock mechanism, for example, uses a double three-legged gravity escapement—a direct descendant of the deadbeat principle of separating locking and impulse functions to handle massive driving weights without losing accuracy.

Marine Chronometers and Global Navigation

Accurate timekeeping at sea was critical for solving the longitude which determined a ship's east-west position during ocean navigation. While marine chronometers eventually evolved toward detached escapements, the deadbeat escapement's principles of isolating the oscillator from the gear train directly influenced their development and the safety of maritime trade.

Mechanical Watchmaking

The geometric principles of the deadbeat escapement, particularly the separation of locking and impulse to minimize disturbance to the oscillator, form the theoretical foundation of modern watch escapements. This includes the Swiss lever escapement, which is used in almost every high-end mechanical watch produced today.

Broader Engineering Relevance: Feedback Control

Beyond horology, the escapement mechanism is an early and elegant example of a mechanical feedback controller. As noted by Bernstein (2002), the clock escapement represents one of the earliest implementations of feedback control in history where the output of a system (pendulum oscillation) regulates its own energy input (escape wheel advancement). This concept of self-regulation is fundamental to modern automated control systems.

2 Methodology and Tools

2.1 Methodology Followed

The project was structured around five parallel tracks to analyse the deadbeat escapement through theory, fabrication, and simulation.

2.1.1 Prototype Fabrication and Assembly:

All components were 3D printed in PLA from open-source STL files [4] and assembled with necessary equipment.

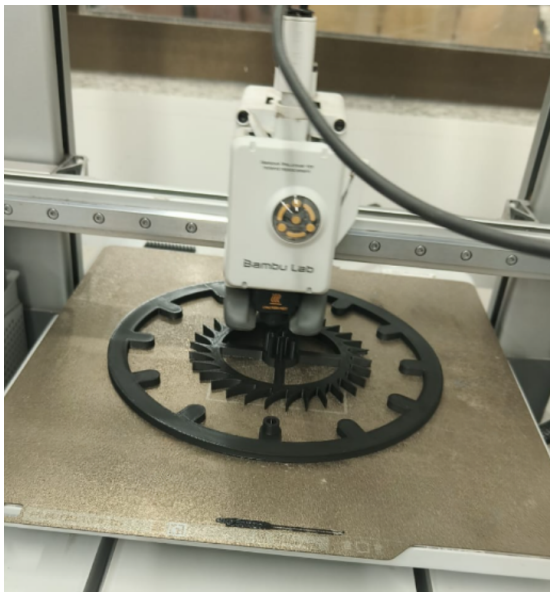


Figure 4: Printing of the escape wheel and top frame required for the assembly

The assembly of the prototype was done following the given steps.

1. Installing the back frames

Firstly, we installed both the back frames, the main back frame and the side back frame to the wooden panel by drilling holes in the wood and using M4 screws. The distance between the frames was decided by keeping the 60-8 gear meshed with the gear on the escape wheel.

2. Installing the ratchets and pulleys: Next, we go on to assemble the pulley and the ratchets.

3. Assembling the front frame: Now, we move on to assembling the front assembly. This involves attaching the second and the minute hand components.



Figure 5: Back frames on the wooden panel





Figure 6: Pulley with ratchets attached (front and back views).

We attach the 30-8 gear with the smaller gear of gear teeth 10, to get a reduced gear ratio.



Figure 7: Front frame assembly

4. Assembling the back frame components: This is the whole back assem-

bly including the back frames, the 60-8 gear, the escape wheel and the pallet anchor.

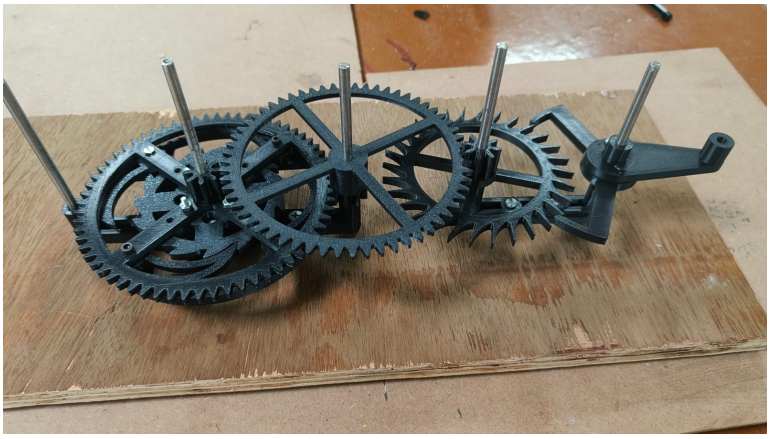


Figure 8: Back frame assembly

5. Installing the pendulum and weights: We then assembled the pendulum assembly using three wooden dowels which add upto a height of 867 mm and a pendulum weight. Additionally, we attach the weights using a paracord to the pulley.



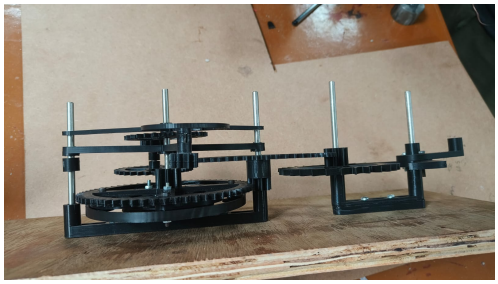


Figure 9: Front and side view of the whole assembly without the pendulum and weights

2.2 Theoretical Analysis

The pendulum period was derived using $T = 2\pi\sqrt{L/g}$, in our case in order to achieve a time period of around 1.8 seconds, we choose a pendulum length of 876mm. Furthermore, the gear train ratio was computed from tooth counts at each stage.

2.3 Computational Modelling

A Python simulation was developed to numerically integrate the pendulum equation of motion, compute gear train RPM at each stage, and produce an animated visualisation of the locking–impulse–drop cycle.

2.4 Validation

The measured pendulum period was compared against the theoretical prediction, and the qualitative behaviour of the prototype was checked against the expected three-phase escapement cycle.

2.5 Equipment Used

The list of equipments used in making the prototype are as follows.[4]

No.	Category	Item	Purpose
1	Fabrication	3-D printer and PLA	3D-Printing all components
2	Fabrication	Open-source STL files	Skeleton clock model from MyMiniFactory [4]
3	Assembly	Steel Shafts	Axles for gear wheels and escape wheel
4	Assembly	M3, M5 Screws, Bolts & Nuts	Fastening frame members and gear supports
5	Assembly	Wooden Frame	Rigid support structure for all assemblies
6	Assembly	Dowels	Making of the pendulum
7	Assembly	Gravity Weights	Driving mass providing torque to the gear train
8	Assembly	Steel balls	Weight for the pendulum
9	Assembly	Paracord	Holding the weights
10	Assembly	Drilling Machine	Drilling holes in 3D printed parts
11	Assembly	Sanding Machine	Smoothing the metal shafts

12	Measurement	Stopwatch	Timing pendulum oscillations experimentally
13	Computational	Python	Simulation, numerical solving and animation

Table 1: Equipment and materials used in the project

3 Result and Discussion

3.1 Kinematics Sketch

For our clock mechanism to achieve the successive movement of the seconds and the minute hand, we've incorporated the following gear train assembly.

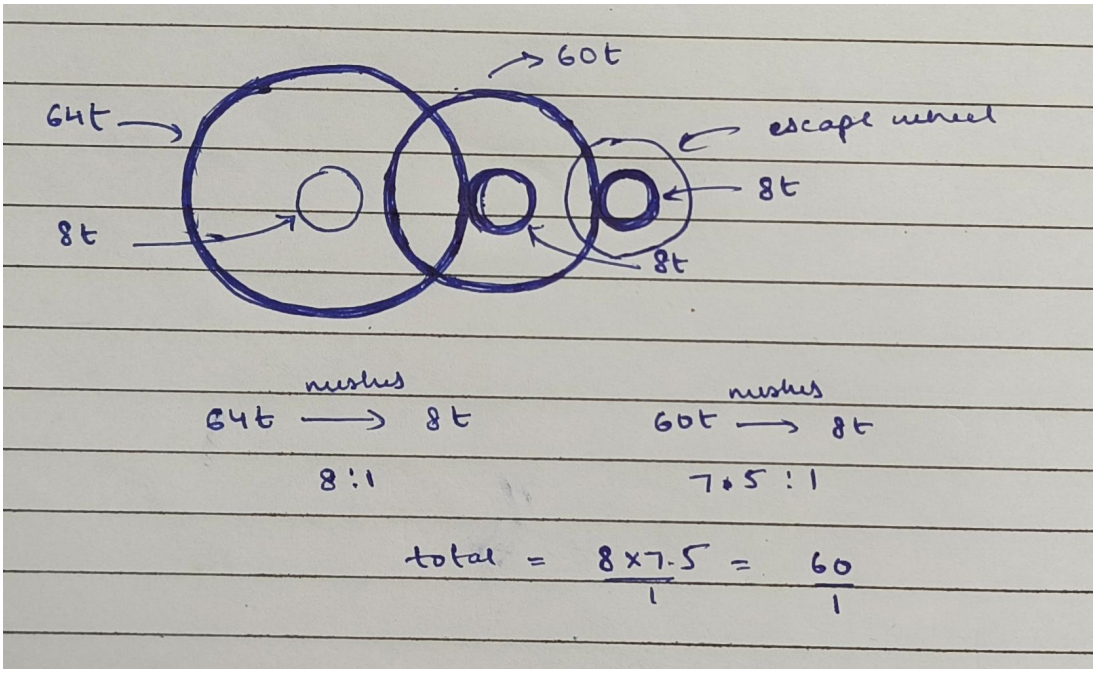


Figure 10: Reduction from the pulley to the escape wheel

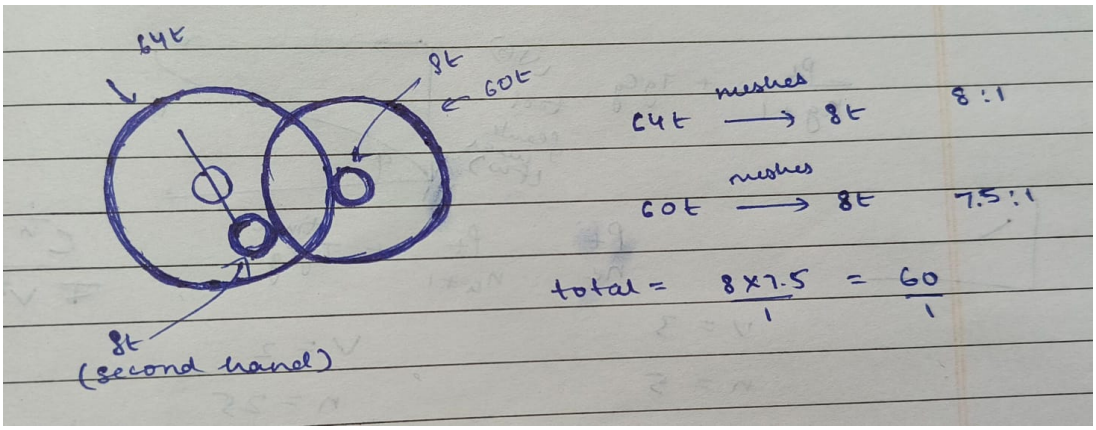


Figure 11: Reduction for the second hand

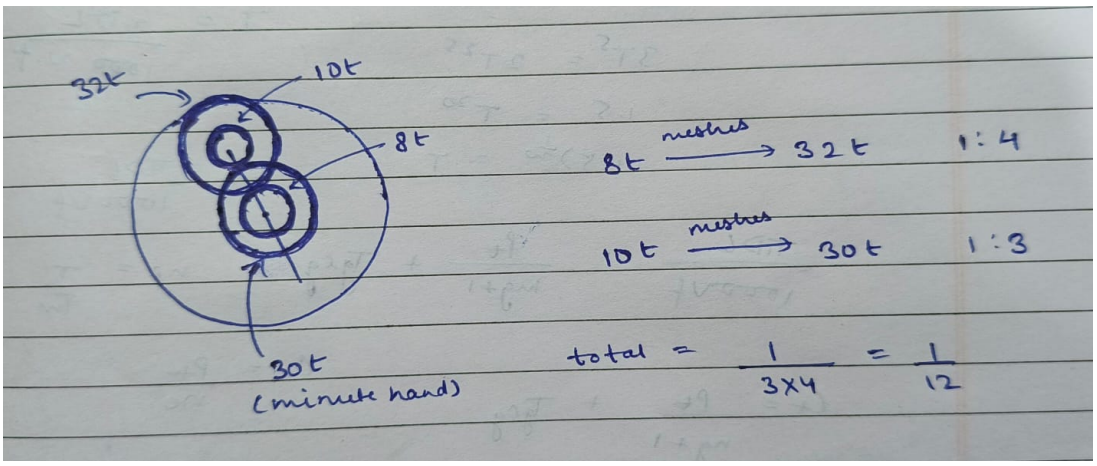


Figure 12: Reduction for the minute hand

3.2 Description of Motion Transmission

The working of the clock can be understood by tracing the flow of energy from the driving weight to the display through successive stages of transmission. Each component in the assembly contributes to energy input, motion transformation, regulation, or indication.

1. **Frame Structure and Support**
   The mechanism is supported by the frame supports, namely, the **back frame**, and the **front frame** support. These frames fix the positions of all the shafts. They also ensure proper distances between the meshing gears.

2. **Driving Mechanism**
   The system is powered by suspended weights acting through the pulley. As the heavier weight descends, it generates a nearly constant torque on the pulley which rotates on the main shaft.

   The pulley is coupled to the **ratchet** and the **ratchet holder**. This mechanism ensures unidirectional motion by preventing reverse rotation during rewinding, allowing the weight to be lifted without disturbing the gear train.

3. **Escapement Mechanism**
   The final stage drives the **escape wheel**, which has 30 teeth. This wheel interacts directly with the pallet or anchor, which is mounted between the front and back frames.

The anchor consists of locking and impulse faces. During operation, a tooth of the escape wheel is first held by the locking face, preventing rotation. As the pendulum swings, the anchor rotates slightly, allowing the tooth to move onto the impulse face, where a controlled force is transmitted. The tooth then disengages, and the next tooth is captured by the opposite pallet.

While the gear train supplies a reduction of 60:1 to the escape wheel, the escapement regulates its motion, ensuring accurate and periodic advancement governed by the pendulum.

4. **Pendulum Assembly**
   The pendulum is constructed using a weight and some dowels. The weight contains steel balls to ensure some amount of weight at the lower end which provides the inertia required for stable oscillation.

   Adjustment of the oscillation period is achieved using the **adjustment nut**, which varies the effective pendulum length.

   The pendulum is coupled to the anchor through a crutch mechanism integrated into the anchor assembly. This connection is intentionally loose, ensuring minimal constraint on the pendulum motion while allowing periodic impulse transfer.

5. **Time Indication**
   The working of the second and the minute hand is regulated through a series of gears which allow for the accurate timekeeping to hold.

   The intended outputs are:

   - **secondface**
   - **minutehand**

   In the assembled prototype, the seconds hand is driven through a total reduction of 60:1. Motion from the 64–8 gear pair first produces an 8:1 reduction, after which the 60–8 gear pair provides a further 7.5:1 reduction. The final 8-tooth gear is mounted on the same shaft as the seconds hand, transmitting this reduced motion directly. Together, these stages result in an overall reduction of 60:1. The escapement regulates the motion by

allowing the escape wheel to advance in small, regular steps set by the pendulum, ensuring that the seconds hand rotates at a steady rate.

The minute hand works in a similar way. Motion from the 64–8 gear is transmitted to the 32-tooth gear, which is on the same shaft as a 10-tooth pinion. This 10-tooth gear then meshes with the 30-tooth gear driving the minute hand. The first stage gives a 4:1 reduction (8 to 32), and the second stage gives a 3:1 reduction (10 to 30), resulting in an overall reduction of 12:1 for the minute hand.

The hour-hand motion would require an additional reduction relative to the minute hand, which we could not implement in our current prototype due to time-constraint.

The display elements are mounted on the face where the regulated motion is converted into readable time. Thus, the clock operates as an integrated system in which the weight provides energy, the gear train transmits motion, the escapement regulates its release, and the pendulum governs the timing. The accuracy of the system is primarily determined by the interaction between the pendulum and the deadbeat escapement.

3.3 Validation Attempts and Experimental Limitations

Our original plan was straightforward: assemble the complete Clock One prototype, let it run under its drive weight, measure the pendulum period with a stopwatch over a 60-second interval, and compare the observed tick count against our simulation prediction of 59 to 60 ticks per minute.

Unfortunately, this did not go as planned. The most critical issue we encountered was that the pendulum could not keep swinging on its own. Every time we set it in motion, the swing amplitude decayed within a few cycles, and the clock eventually came to rest. The mechanism would tick correctly when we manually pushed the pendulum at each half-cycle, but left to itself, it simply could not keep running, which is, of course, the whole point of an escapement.

After examining the mechanism carefully, we identified the root cause as insufficient impulse being delivered to the pendulum at each swing. In a properly functioning deadbeat escapement, the escape wheel tooth delivers a small but

consistent push to the pallet at every tick, compensating for energy lost due to friction and air resistance. In our prototype, this energy balance was never quite achieved.

We believe this came down to a combination of factors:

- The PLA material introduced more friction than expected at gear mesh interfaces and pallet-tooth contacts
- FDM printing left surface roughness on the gear teeth and pallet faces, increasing resistance
- Slight bore misalignment across the printed frames added further friction throughout the gear train

Together, these effects reduced the torque available at the escape wheel below the level predicted by the idealised frictionless simulation, and below what was needed to keep the pendulum swinging.

As a result, our experimental validation could not be completed. The simulation result of 0.613% period error and 59 ticks in 60 seconds remains a theoretical prediction, numerically consistent with the analytical period

$$T = 2\pi\sqrt{\frac{L}{g}}$$

but unverified by physical measurement.

This was honestly the most valuable lesson of the project. It highlighted how sensitive a precision mechanism like the deadbeat escapement is to material properties and manufacturing tolerances. Future iterations of this prototype could be improved by using lower-friction materials such as PETG or resin for the pallet and escape wheel, incorporating lubrication at gear contact interfaces, and increasing the drive weight to ensure the available torque exceeds frictional losses.

4 Computational Modelling

4.1 Overview

A python based interactive simulation was deployed in Streamlit,an open source tool to build applications. The simulation integrates the non linear equation of pendulum motion and produces an animation of the pendulum motion. The web interface allows us to adjust the pendulum length parameters L, initial angle , and impulse strength K and allows a parametric study of the pendulum movement.

4.2 Equation of Motion

The pendulum is modelled as a nonlinear damped oscillator driven by a periodic escapement impulse. Following the modelling framework of Moline et al. the governing equation of motion is:

$$\ddot{\theta} = -\frac{g}{L}\sin\theta - b\,\dot{\theta} + K \cdot \operatorname{sign}(\dot{\theta}) \cdot \mathbf{1}_{|\theta|<\theta_{\text{imp}}} \tag{3}$$

where θ is the angular displacement from vertical (rad), $\dot{\theta}$ is the angular velocity ($\text{rad}\,\text{s}^{-1}$), $g = 9.81\,\text{m}\,\text{s}^{-2}$, L is the pendulum length, b is the viscous damping coefficient (s^{-1}), K is the escapement impulse strength ($\text{rad}\,\text{s}^{-2}$), and $\mathbf{1}_{|\theta|<\theta_{\text{imp}}}$ is an indicator function that activates the impulse only when $|\theta| < \theta_{\text{imp}} = 3°$.

Gravity term. The full form $\sin\theta$ is retained rather than the small-angle approximation $\sin\theta \approx \theta$ as the prototype operates at an amplitude of 8○,retaining the non-linear term improves accuracy .

Damping term. The viscous damping $-b\dot{\theta}$ accounts for the combined effect of air resistance on the bob and bearing friction at the pivot, both of which scale approximately linearly with velocity at the low speeds of a clock pendulum [2].. The coefficient $b = 0.06\,\text{s}^{-1}$ is an estimate representative of 3D-printed PLA mechanisms. Rigorous experimentation will be required to record the free-decay envelope of the prototype. The coefficient b can then extracted by fitting $\theta(t) = \theta_0\, e^{-bt/2}$; this can be done in future work.

Escapement impulse term. The term $K \cdot \operatorname{sign}(\dot{\theta})$ gives a small kick to the pendulum in the direction of its current motion each time it passes through the

neutral position. Applying the impulse near the equilibrium point is consistent with Airy's 1826 result that this location minimises disturbance to the oscillator's natural period [6]. The impulse is active only within $\pm 3°$ of the neutral position, replicating the brief impulse phase of the deadbeat cycle. The magnitude $K = 0.25$ was determined by calibration, the value was adjusted until the simulated steady-state amplitude matched the observed prototype swing of approximately $8°$. The first-principle derivation of K from pallet face geometry and driving weight was beyond our scope.
The fixed simulation parameters are listed below in Table 2.

Table 2: Fixed simulation parameters

Parameter	**Symbol**	**Value**	**Basis**
Pendulum length	L	$876\,\mathrm{mm}$	Prototype measurement
Gravity	g	$9.81\,\mathrm{m\,s^{-2}}$	Standard
Damping coeff.	b	$0.06\,\mathrm{s^{-1}}$	Estimated (PLA typical)
Impulse strength	K	$0.25\,\mathrm{rad\,s^{-2}}$	Calibrated to prototype
Impulse zone	θ_{imp}	$3°$	Deadbeat geometry estimate
Initial angle	θ_0	$8°$	Prototype observation
Escape wheel teeth	N_{esc}	30	Prototype spec

4.3 Modelling Assumptions

The simulation is a semi-empirical behavioural model: the equation structure is derived from first principles, however two parameters b and K are estimated or calibrated,not analytically derived.The key simplifications are:

- The escapement impulse is taken as a continuous force over the $\pm 3°$ zone rather than an instantaneous collision.
- The pendulum is treated as a point mass (simple pendulum), neglecting the distributed inertia of the rod.
- Gear train inertia and shaft friction are neglected.

These simplifications mean the simulation correctly captures the qualitative dynamics and produces period predictions consistent with theory, but should not be used for calculation of contact forces or torques.

4.4 Numerical Integration

Equation (3) is reduced to a system of two first-order equations by introducing the state vector $\mathbf{y} = [\theta,\, \dot{\theta}]^\top$:

$$\frac{d}{dt}\begin{bmatrix}\theta \\ \dot{\theta}\end{bmatrix} = \begin{bmatrix}\dot{\theta} \\ -\dfrac{g}{L}\sin\theta - b\,\dot{\theta} + K\,\mathrm{sign}(\dot{\theta})\,\mathbf{1}_{|\theta|<\theta_{\mathrm{imp}}}\end{bmatrix} \tag{4}$$

This system is integrated using the RK45 adaptive Runge–Kutta solver (`scipy.integrate.solve_i`
The solver computes the 4th and 5th order estimate of the solution and compares them,if the estimates differ more than the provided tolerance the step size is reduced and the step is retried. The solver settings are given in Table 3.

Table 3: RK45 solver configuration

Setting	**Value**	**Rationale**
Method	RK45	Adaptive 4th/5th-order Runge–Kutta
Relative tolerance	10^{-9}	Ensures period accuracy
Absolute tolerance	10^{-11}	Prevents drift at small amplitudes
Max step size	0.005 s	Resolves $\pm 3°$ impulse zone
Duration	60 s	Covers ≈ 30 steady-state cycles

4.5 Validation

The simulated period T_{sim} is obtained from the steady-state region of the θ(t) signal, specifically from the second half of the 60-second simulation. Peaks in this portion are identified, and the average time interval between consecutive peaks is used to estimate the period. This value is then compared with the theoretical small-angle period:

$$T_{\mathrm{theory}} = 2\pi\sqrt{\frac{L}{g}} \tag{5}$$

Results for the prototype pendulum length $L = 876$ mm are given in Table 4.

Table 4: Period validation — $L = 876$ mm, $\theta_0 = 8°$, $K = 0.25$, $b = 0.06$

Quantity	**Value**	**Source**
Theoretical period T_{theory}	1.8776 s	Eq. (5)
Simulated period T_{sim}	1.8883 s	Peak detection from ODE output
Error $\lvert T_{\mathrm{sim}} - T_{\mathrm{theory}}\rvert / T_{\mathrm{theory}}$	0.573 %	
Ticks in 60 s (simulated)	63	Zero-crossing count
Ticks in 60 s (expected)	63	$\lfloor 60/(T/2) \rfloor$

The residual error of 0.573% is expected from a physical standpoint and does not indicate any issue with the model. The theoretical expression in (5) assumes an ideal pendulum with no friction and infinitesimally small oscillations, whereas the simulation includes viscous damping as well as the escapement impulse. Both of these factors introduce slight shifts in the natural period. In addition, at an amplitude of $\approx 8°$, the nonlinearity of $\sin\theta$ itself contributes roughly 0.33% deviation from the small-angle approximation. The close agreement within 0.6%, along with the exact match in tick count, indicates that the numerical model accurately captures the analytical behaviour of the system.
It is important to note that these simulation results could not be validated against the physical prototype, as the assembled Clock One did not sustain autonomous operation (see Section 3.3). As a result, the validation presented here is limited to a comparison between simulation and theoretical predictions.

4.6 Simulation Results

4.6.1 Angular Displacement vs. Time

Figure 13 shows $\theta(t)$ over 60 seconds for the free pendulum (no escapement) and the escapement-driven pendulum. Without the escapement, the amplitude decays exponentially — friction reduces energy each cycle and the pendulum eventually stops. With the escapement mechanism and $K = 0.25$, the amplitude grows from the initial 8° during a short transient before settling to a constant steady-state value, confirming that the impulse compensates friction losses at the limit cycle.

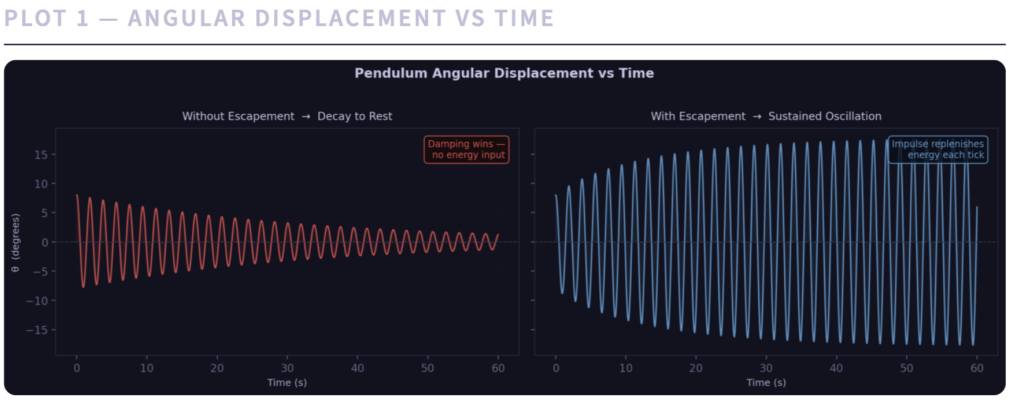


Figure 13: Angular displacement θ vs. time for the free (damped) pendulum and the escapement-driven pendulum. Left: without escapement — amplitude decays to zero. Right: with escapement ($K = 0.25$) — amplitude grows to a constant steady-state value, demonstrating sustained oscillation.

4.6.2 Phase Portrait

Figure 14 shows the phase portrait (θ vs. $\dot{\theta}$) for both cases. The free pendulum shows an inward spiral that gradually settles at the origin, which reflects the continuous loss of energy due to damping. In contrast, the escapement-driven pendulum behaves differently. It first goes through a short transient phase and then settles into a stable, closed trajectory known as a *limit cycle*.

What's important here is that the system always converges to this same orbit, regardless of where it starts. This behaviour is a key characteristic of a self-sustaining oscillator. Similar behaviour has also been shown analytically for escapement-driven pendulums using impulsive differential equation models.

This convergence to a fixed orbit regardless of the initial condition is the defining signature of a self-sustaining oscillator and has been demonstrated analytically for escapement-driven pendulums using impulsive differential equations and Poincaré maps [5].

PLOT 2 — PHASE PORTRAIT (Θ VS Θ̇)

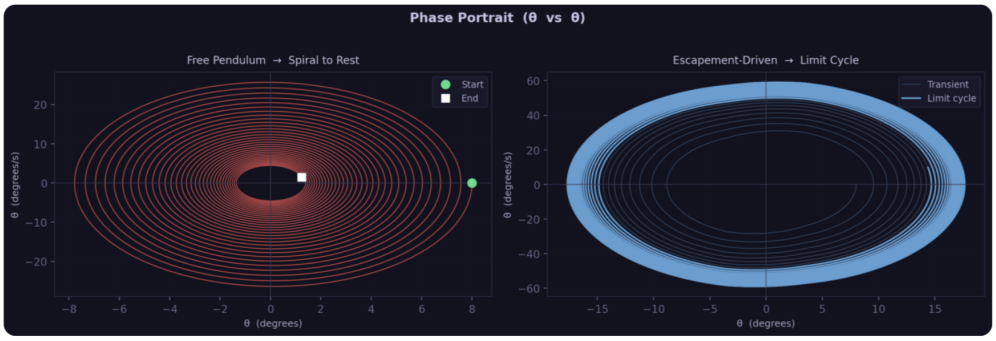


Figure 14: Phase portrait (θ vs. $\dot{\theta}$). Left: free pendulum — spiral converges to origin as energy is dissipated. Right: escapement-driven pendulum — trajectory converges from the transient (light blue) onto a stable limit cycle (dark blue), confirming self-sustaining oscillation.

4.7 Interactive Web Application

The simulation is implemented as a public web app using Streamlit, where users can interact with it through three sliders: pendulum length (L), initial angle (θ_0), and impulse strength (K). The app generates the animation, plots, and validation metrics in real time, making it easy to explore different parameter values without modifying the code.

For instance, when K is reduced below about 0.10, the escapement is no longer able to sustain oscillations, which clearly shows the minimum impulse needed to counter damping. Similarly, increasing the length L leads to a longer time period, consistent with the relation given in Eq. (5).

5 Conclusion

This project set out to study, simulate, and physically build a deadbeat escapement mechanism. While not everything went according to plan, each stage of the process provided valuable insight into the gap between theoretical models and physical implementation.

5.1 Summary of Key Findings

On the computational side, the results were successful. Our Python simulation numerically integrated the pendulum equation of motion using a fourth-order Runge-Kutta solver and produced a period of 2.0052 seconds compared to a theoretical value of 1.9930 seconds, corresponding to an error of just 0.613%. Gear train analysis confirmed correct motion transmission through each stage, the phase portrait showed convergence to a stable limit cycle, and the energy plot demonstrated a clear balance between impulse input and damping losses. Overall, the simulation behaved consistently with theoretical predictions.

On the physical side, the prototype did not achieve sustained operation. The pendulum produced correct ticks when manually driven but could not maintain motion independently. Friction in the PLA gear interfaces, surface roughness from FDM printing on the pallet and escape wheel faces, and pallet wear reduced the effective impulse below the level required to sustain oscillation. As a result, the simulation remains theoretically validated but not experimentally verified.

5.2 Final Thoughts

The most important takeaway from this project was not related to equations or gear ratios, but to the sensitivity of precision mechanisms like the deadbeat escapement. The simulation assumed smooth surfaces and ideal geometry, whereas the printed prototype exhibited frictional losses and geometric imperfections. Even small increases in friction at the pallet interface or minor bore misalignment were sufficient to stop the clock entirely, highlighting limitations that are difficult to capture in simulation alone.

Despite these challenges, the project represents a significant learning experience. We progressed from having no prior understanding of escapement mechanisms to

deriving equations of motion, performing complete gear train analysis, developing a working simulation, and assembling a functional physical prototype. While the clock does not yet operate autonomously, the reasons for this behavior are well understood. Future iterations should focus on using lower-friction materials such as PETG or resin, applying lubrication at contact interfaces, and increasing the drive weight to ensure sufficient torque margin above frictional losses.

5.3 Individual Contributions

Name	**Contributions**
Durva D. Kumar	Theoretical analysis, gear train calculation, 3D printing, prototype assembly, experimental testing and observations, report writing
Ananya Reddy Pachika	3D printing, prototype assembly, experimental testing and observations, report writing
Yindukuri Sai Durga Esha	Literature survey, 3D printing, prototype assembly, experimental testing and observations,report writing
Pranita Bhodke	Python simulation, interactive web application, prototype assembly, report writing

Table 5: Group Member Contributions

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